



UNIVERSITY OF AMSTERDAM

Tutorial 2 / Part 1:

Scraping: Collecting Web Data for Marketing Analytics

Digital Marketing and Analytics: Week 1 Tutorial 2 (Part 1)

Academic Year 2025-2026 – Semester 1 Block 2

AMSTERDAM
BUSINESS
SCHOOL

Business

Collecting Web Data: Scraping

Web Scraping: Collecting On-Site Web Data

Collecting data from web: Social media posts, Product listings, Customer reviews, ratings, comments, using scraper-software
Importing to your local data base in xls, json, sql or other format.

*This data is **the data you SEE on the website**. NOT the click-stream data (i.e. GA-4) which you cannot directly observe.*

Scrapers: Today used by most firms, researchers and analytics agencies.



Web Scraping: Common Question – Is it Legal?

Mostly YES. If PUBLICLY AVAILABLE data and if used for analytical-research-educational purposes.

However If you INTEND to commercialize scraped data by reproducing, selling, re-selling to third parties it can well be Illegal.

Many firms allow scraping (conditionally) : Twitter, Booking.com, YELP, Amazon, IMDB, VIVINO etc.*

For others, this is still a grey area or permission needed: Facebook, Instagram, LinkedIn

Note: Sometimes firms change-adjust their policy on scraping. See Example from TWITTER in 2023. If you are interested in Scraping you better follow these too.



Elon Musk 
@elonmusk

Subscribe



To address extreme levels of data scraping & system manipulation, we've applied the following temporary limits:

- Verified accounts are limited to reading 6000 posts/day
- Unverified accounts to 600 posts/day
- New unverified accounts to 300/day

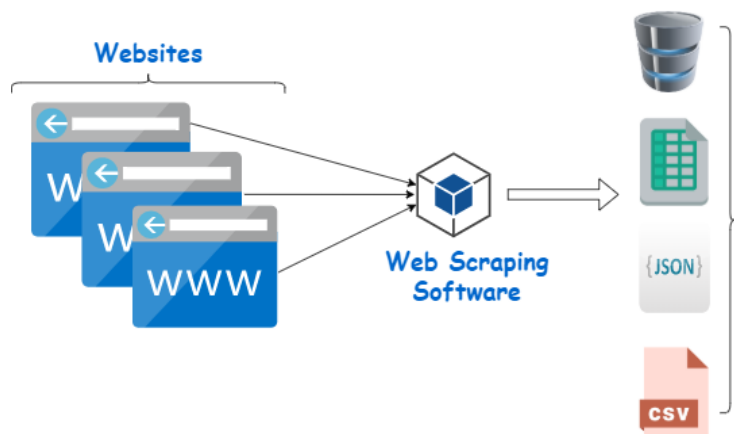
1:01 PM · Jul 1, 2023 · 533.1M Views

Scraping: Collecting (On-Site) Web Data

Web Scraper will only give you data **that you see on the website available and visible to visitor**: product features, ratings, reviews, etc.

Does **NOT** give you **what visitors do on that website, what and how they click !** (non-visible information)

它不会提供访客在网站上的具体操作，例如他们点击了什么以及如何点击！（这些信息不可见）



Scraping Tools: Basic & Free Software

1 Browser Extensions: Examples: Data Miner & Webscraper

PROS: Fastest solution for simple-small data sets. Easy to install and learn. Free.

CONS: Not sufficient for complex tasks. Crashes or Malfunctions often depending on browser or operating system settings.

优点：处理简单小型数据集速度最快。易于安装和学习。免费。

缺点：不足以应对复杂任务。经常崩溃或出现故障，具体取决于浏览器或操作系统设置。

2 Scraping Software-Services: Examples: Parsehub & Octoparse

PROS: User-Friendly visual dashboards. Fast and efficient. Used by many firms. Usually have free plans for academics and small data sets-projects.

CONS: Subscription-Payment needed for very large projects. Although good enough may not suit for very complex-customized tasks.

优点：用户友好的可视化仪表盘。快速高效。众多企业都在使用。通常为学术界和小数据集项目提供免费方案。

缺点：大型项目需要订阅付费。虽然功能足够强大，但可能不适合非常复杂或定制化的任务。

CHROME/ BROWSER EXTENSIONS

SCRAPING SOFTWARE SERVICES



Scraping Tools: Complex & Large Data Sets & Enterprise Solutions

For very large data sets (millions of pages) OR complex tasks (multiple pages, multi-layered scheduled tasks)

3 R or Python Scraping Libraries-Kits:

优点：功能强大、可定制，针对极其复杂的大型任务提供高级解决方案，几乎没有任何限制，完全免费。
缺点：需要一定的编程知识和较多的学习时间。通常并非商业软件，因此用户需要自行完成许多工作。

PROS: Sophisticated, Customizable, Advanced solutions for very complex-large tasks with almost no limits. Absolutely Free.

CONS: Some coding knowledge and more learning time needed. Often not a commercial software thus the user needs to do many things on own.

R Scraping Libraries: Rvest, RCrawler, RSelenium



rvest

Python Scraping Libraries: Scrapy, Selenium, BeautifulSoup



BeautifulSoup



Scraping Tools: What Do We Learn in This Course?

Browser Extensions (Dataminer, Webscraper)

Optional – See Appendix & Video Clips

Easy to learn on your own and fast-handly for small data sets. However often may crash or cause problems depending on browser, webpage, keyboard-key settings and cause complications. Therefore I don't take extensions as the basis – though for those who want to learn I have added a DataMiner Appendix in the end.



SCRAPING SOFTWARE-SERVICE

As one of the most widely used and easy to use scraping-software services with a visual-good dashboard we take **PARSEHUB** as our main platform to practice in this course. PARSEHUB is free for academic use and small projects

R & Python (Rvest & Tweepy)

Optional – See the slides & Video Clips

Probably the best domain to learn-be an expert in scraping. However extensive and takes time.

Therefore I leave this optional too. Though I would like you to have a smooth-practical start with these so you can (if you want) take it further.

We show you two simple examples with R (Rvest – IMDB scraping) and Python (Tweepy – a specialized library for Twitter scraping)

Scraping Web Data

Scraping List Format - Three Main Steps

Scraping Basics: Data in List Format

Good to know before you start !

Most data that we may wish to scrape is in **LIST format**: Thus in a STRUCTURED pattern – repeating itself: check the examples below !

Therefore it is important to detect the list the elements/components of a list.



Tallulah101
York, United Kingdom
62 17

Reviewed yesterday via mobile

Great food and great service

Visited with a group of friends including 2 vegetarians . Great selection of food for all of us and all was delicious , the baked cauliflower that we all had was amazing , the selection of cocktails were many and enjoyable and if you like... [More](#)

Thank Tallulah101



Mike5469
Durham, United Kingdom
7 6

Reviewed 3 days ago

Please give it a go.

The only problem with this restaurant is that it looks like just a normal Pub! Don't be fooled, please give the food a go, as it is really good. It's the second time we have visited and the menu had changed but was even better.... [More](#)

Thank Mike5469



Simon H
7 5

Reviewed 1 week ago via mobile

Interesting

Really great meal, with very different food, which every dish was tasty, notably the oxtail lasagne! Service was friendly and enthusiastic. Only reason this isn't a 5 star review is in an attempt to make the whole experience quirky the seating is awful, we were... [More](#)

Thank Simon H



Apple iPhone X 64GB - GSM & CDMA Unlocked -USA Model -Apple Warranty -BRAND NEW!
GSM+CDMA UNLOCKED × APPLE WARRANTY × FREE SHIPPING × Brand New

\$989.00
Buy It Now
Free Shipping
4403+ Sold

Guaranteed by **Fri, Apr. 13**
Top-rated seller



Apple iPhone X 256GB - GSM&CDMA Unlocked-USA Model-Apple Warranty- BRAND NEW!
GSM+CDMA UNLOCKED × APPLE WARRANTY × FREE SHIPPING × Brand New

\$1,269.99
Buy It Now
Free Shipping
5912+ Sold

Guaranteed by **Fri, Apr. 13**
Top-rated seller



NEW LISTING Apple iPhone X - 64GB - Silver (1-Mobile) A1901 (GSM). Unopened. Brand New
Brand New
★★★★★ 21 product ratings

\$950.00
Trending at \$999.49
or Best Offer
+\$9.85 shipping
18 new & refurbished from \$880.00

Scraping: Three Main Steps – *and we always have these three steps*

No matter which basic scraping tool you use there are three logical steps in scraping. The main logic is all the same for all softwares/packages.

- (1) Define the **CONTAINER (main unit/parent)** that makes up the data (series) on website. This is also the **ROW** structure of your data.
- (2) Define the **SUB-ELEMENTS (child)** in the container, you wish to scrape data about (price, rating, name, etc) each of them makes each column in your data.
- (3) Define the **PAGINATION (next page: function)** so scraper can automatically “click” and go the subsequent pages and keeps scraping.

The screenshot shows a Yelp business page for 'The Bird' in Washington, DC. Red boxes and arrows highlight various elements for scraping:

- Business Name:** The Bird
- Claimed Status:** claimed
- Number of Reviews:** 35 reviews
- Category:** American (New), Breakfast & Brunch
- Price Range:** \$\$\$
- Address:** 1337 11th St NW, Washington, DC 20001
- Phone Number:** (202) 518-3609
- Website:** thebirddc.com
- Hours:** A table showing working hours for each day of the week.
- More business info:** A list of additional details about the business.

Day	Hours
Mon	4:00 pm - 10:30 pm
Tue	4:00 pm - 10:30 pm
Wed	4:00 pm - 10:30 pm
Thu	4:00 pm - 10:30 pm
Fri	4:00 pm - 11:30 pm
Sat	10:00 am - 11:30 pm
Sun	10:00 am - 10:30 pm

Takes Reservations	Yes
Delivery	No
Take-out	Yes
Accepts Credit Cards	Yes
Accepts Android Pay	No
Good For	Dinner
Parking	Street
Bike Parking	Yes
Wheelchair Accessible	Yes
Good for Kids	No
Good for Groups	Yes
Attire	Casual
Ambience	Trendy
Noise Level	Average
Alcohol	Full Bar

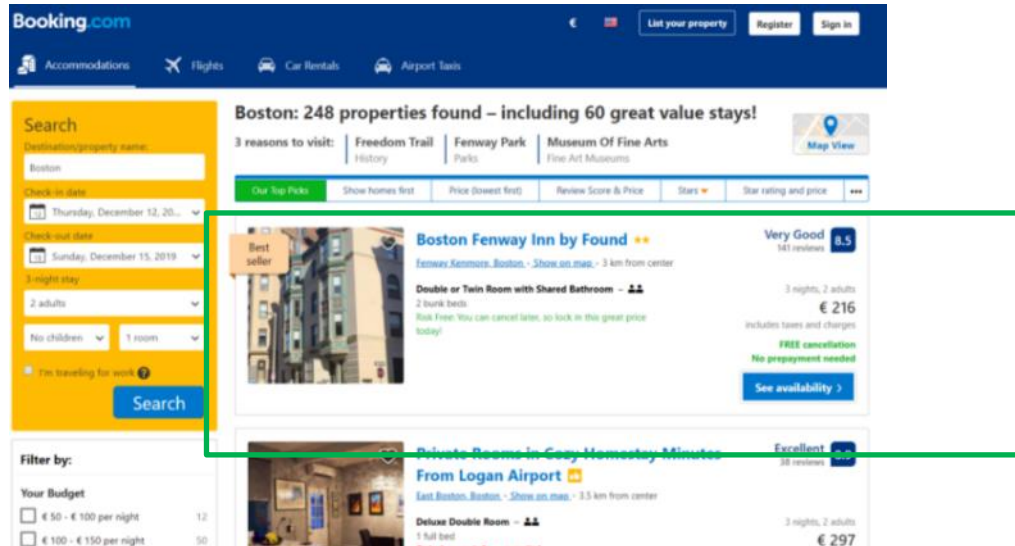
Scraping: Step 1 – Container/Main Unit *(You are collecting data about what?)*

CONTAINER: the main unit in your data: *Let's say we scrape data on all hotels in Boston from Booking.com*

KEY QUESTION: *I am collecting data about what – (here) HOTELS – each hotel with all relevant information*

Then the main unit of your data will be **each box/area** with regard to **each HOTEL** (repeating hotel-by-hotel until the last one) right?

*That will also make the **ROWS** in your final data: Each hotel and its information: one row in your final data set.*



STEP 1:

Choose & Define the Container. The main unit in your data.
Here it is each HOTEL right ?

Scraping: Step 2 - Sub-Elements *(What Exact Information You Need about Each Hotel?)*

What do you want to know about each hotel ? Now we need to identify sub-elements in each container that we shall scrape information about

These could be hotel name, price, score rating, cancellation policy etc. right ? You name it ! It is up to what you want to analyze.

Each of this information element will make another **column** in your scraped data.

STEP 2: Mark/Identify them one by one (the ones that you wish to have in your data) !

Our Top Picks Show homes first Price (lowest first) Review Score & Price Stars ⚑ Star rating and price ...

Best seller

Boston Fenway Inn by Found ★★

Fenway Kenmore, Boston · [Show on map](#) · 3 km from center

Double or Twin Room with Shared Bathroom – 2 people

2 bunk beds

Risk Free: You can cancel later, so lock in this great price today!

Very Good 8.5 141 reviews

3 nights, 2 adults

€ 216

includes taxes and charges

FREE cancellation

No prepayment needed

[See availability >](#)

STEP 2:

Choose & Show the Scraper:

What you want to know about each hotel

Scraping: Step 3 Pagination/Going to Next Page

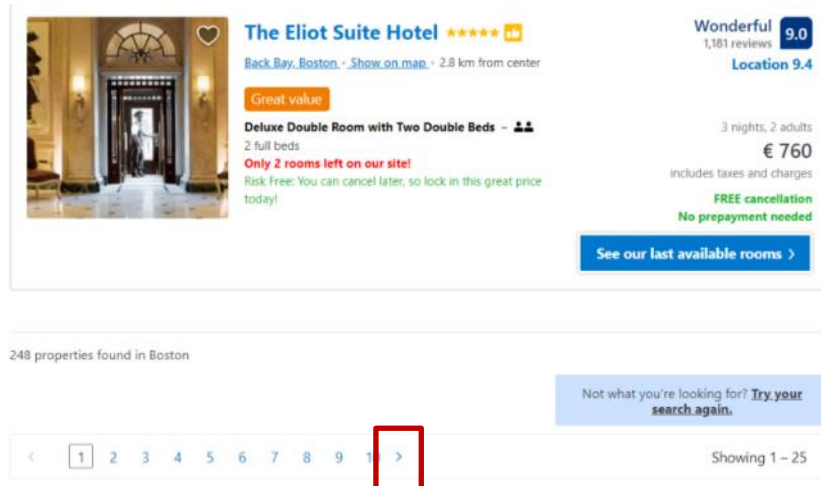
STEP 3: Tell the scraper to click to next page – when it is done with scraping each page.

Booking.com lists in total **243 hotels** in Boston. HOWEVER: on each page there are 30 hotels listed, and those are spread over 9 pages.

Thus: We need to tell/teach the scraper **to automatically click-go to next page and continue scraping (pagination)** until all the data is scraped.

Final step: Mark/Identify the “next page” functionality/button to scraper. So it can do this work for you.

In other words: You are instructing your scraper NOT to stop when it comes to the end of each page – instead click the arrow and continue with the next ones



STEP 3:

Show scraper where “next page” button is
So it can keep scraping after each page is done

Scraping w/ Parsehub

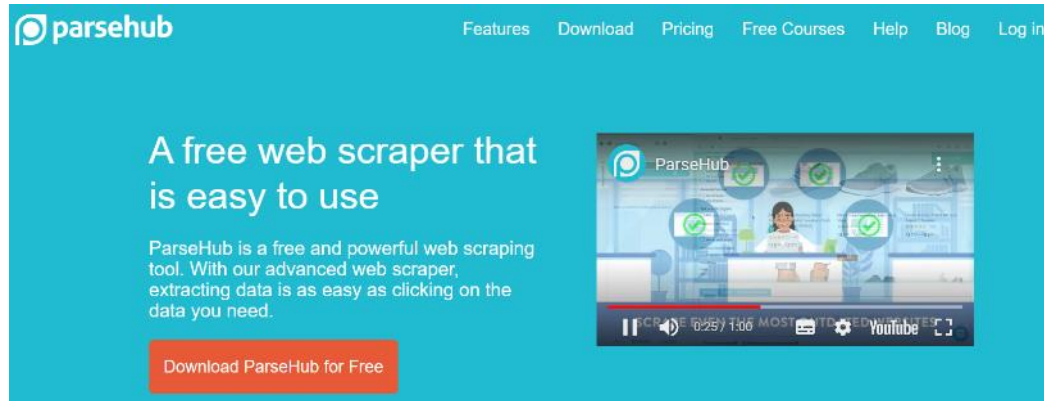
Scraping Software Example (w/ Free Plan)

Scraping w/ Parsehub: General Information

PARSEHUB is an online web-scraping software service. It is one of the most popular scraping dashboards worldwide.

It has a visual and easy to use interface allows you to set your scraping-project by clicking and drag-dropping elements on the screen.

Although being a commercial-subscription based PARSEHUB has a **limited free plan** (perfectly sufficient for this course)



The screenshot shows the ParseHub website with a teal header containing the logo and navigation links: Features, Download, Pricing, Free Courses, Help, Blog, and Log in. The main content area has a teal background with the text 'A free web scraper that is easy to use'. Below this, it states 'ParseHub is a free and powerful web scraping tool. With our advanced web scraper, extracting data is as easy as clicking on the data you need.' A red button labeled 'Download ParseHub for Free' is positioned at the bottom left. On the right, there is a video player showing a woman using the ParseHub interface, with a play button and a progress bar at the bottom.

Everyone	Standard
FREE No credit card required - \$99 value	\$189 per month, cancel anytime
Get 200 pages of data in only 40 minutes	Get 200 pages of data in only 10 minutes
200 pages per run	10,000 pages per run
5 public projects *	20 private projects
Limited support	Standard support
Data retention for 14 days	Data retention for 14 days
-	Save images & files to DropBox or S3

Scraping w/ Parsehub: Main (Parent) Element – What Data: “Movies”

Naturally the first thing to set in our scraping project (or data) is to set what data (main element) we deal with> This is a data about MOVIES. Then MOVIE NAME is a good choice as a main (parent) element in our data. Go to SELECT command and CLICK the title (label) of a movie in the list. You will see it is marked in color. REPEAT (confirm) this for another movie title (label) you will see it is – and now all movie labels in the list are marked-colored as the first two.

Parsehub interface showing the IMDb Top 250 movies page. The left sidebar shows the 'main_template' with 'Select page' and 'Select MOVIE (50)' commands. The main area shows the IMDb search results for 'Top 250' movies, sorted by 'User Rating'. The first two movies, 'The Shawshank Redemption' and 'The Godfather', are highlighted with green and blue borders respectively, indicating they have been selected as the main element.

NOW – PARSEHUB knows the Movie Title will be the main element in your data.

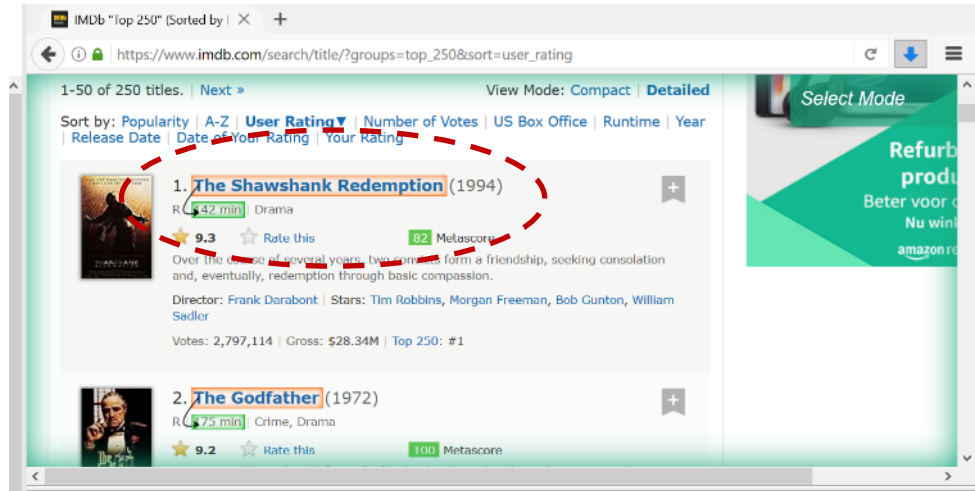
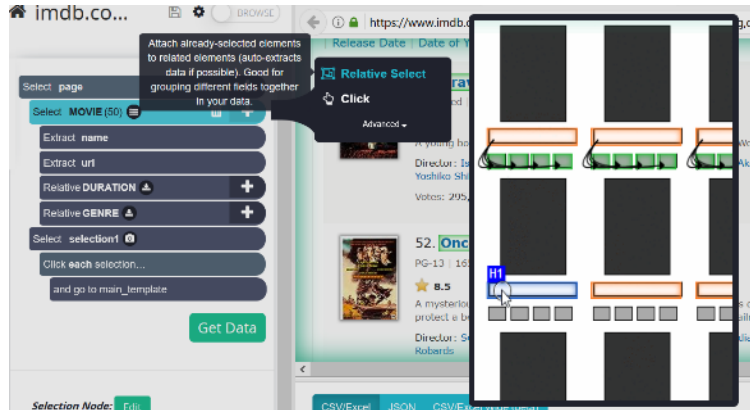
NEXT Let's see how we select-add other elements (pieces of information) about each movie to our data.

We do this by using RELATIVE SELECT command. See Next Slide (how we use it)

Scraping w/ Parsehub: Sub (Child) Elements – What Exactly on Movies?

Now we choose-set what exactly we want about each movie. Let's say we want a data with (i) Movie Name (ii) Movie Duration and (iii) Movie Genre

MOVIE NAME was already selected as (main-parent) element in the previous step. Now we need to link Duration and Genre to each movie. RELATIVE SELECT command simply links your main selection (MOVIE NAME) <=> with (whatever you want to link-connect: here DURATION). After choosing RELATIVE SELECT for your main element (MOVIE NAME) – first click MOVIE NAME and then click DURATION – you will see that PARSEHUB will connect these two with an arrow. Confirm/Repeat this selection-linking with another movie. DONE. DURATION is now added to your data.



Scraping w/ Parsehub: Pagination (Next Page Button)

Now use SELECT again > (w/ your mouse) Choose/Click "Next Page >>" button
Beneath the page so > this tells your scraper to click > GO > next page when it comes to the end of each movie listing page (with 50 movies)
Remember: Our list has 250 movies spreading over 5 pages.

Choose (CLICK) command for your selection "Next Page >>"
PARSEHUB asks you – if what you click is a click to
1 Next Page (pagination) OR
2 Click to go into details of each movie
You choose 1 since you do this to go to the next page

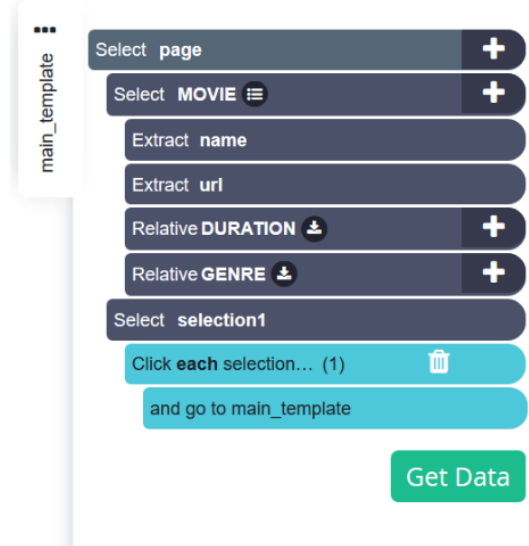
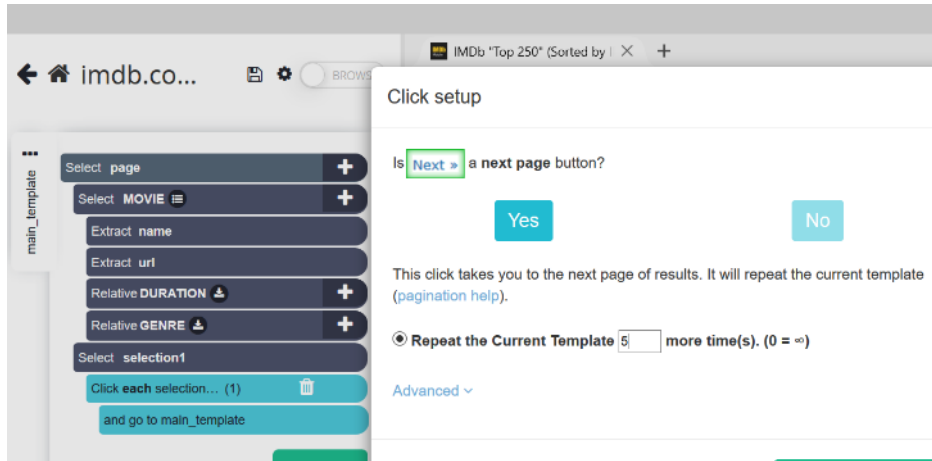
The screenshot shows the Parsehub interface for the IMDb "Top 250" page. On the left, the "main_template" sidebar contains a list of extraction rules: "Select page", "Select MOVIE", "Extract name", "Extract url", "Relative DURATION", and "Relative GENRE". Each rule has a plus icon to its right. Below these rules is a "Get Data" button. On the right, the main content area displays the IMDb page for "The Intouchables" (50th on the list). It shows the movie's title, director (Michael Curtis), votes (591,453), and a brief description. At the bottom of the page, there is a "Next" button for pagination.

The screenshot shows the Parsehub interface with a "Click setup" dialog box open. The dialog asks "Is 'Next >>' a next page button?" with "Yes" and "No" buttons. Below the question, there are two sections: "Examples of next page buttons" and "Examples of non next page buttons". The "Examples of next page buttons" section shows a pagination control with a "Next" button highlighted. The "Examples of non next page buttons" section shows a search bar and a "Search" button. The "Click setup" dialog is circled with a red dashed line.

Scraping w/ Parsehub: Pagination (Next Page Button) & Review Before Running

Now you CONFIRM That the Next Page >> (ARROW) is a next page button – so that your scraper can go until the end of 5 pages list of 250 movies – as each page consist only 50 pages. Your Scraping Project is now ready to RUN.

Before running the scraper;
REVIEW the steps-elements of your
scraping project once again below.

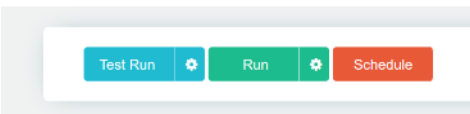


Scraping w/ Parsehub: Downloading your Data

After the scraping process (RUN) ends, PARSEHUB will allow you to download your data in CSV, XLS or JSON formats.

Here is how the data looks in initial CSV format.

← 🏠 [imdb.com](#) Project



Your data is ready! Click on the green buttons to download.



All dates and times are in UTC +0000.

Empty file with no results? [Click here](#) to fix.

CSV file too big? Save the JSON file and [click here to convert to CSV](#).

```

1 MOVIE_name,"MOVIE_url","MOVIE_DURATION","MOVIE_GENRE"
2 The Shawshank Redemption,"https://www.imdb.com/title/tt0111161/?ref=adv_li_tt","142 min","Drama"
3 The Godfather,"https://www.imdb.com/title/tt0068646/?ref=adv_li_tt","175 min","Crime, Drama"
4 The Dark Knight,"https://www.imdb.com/title/tt0468569/?ref=adv_li_tt","152 min","Action, Crime, Drama"
5 Schindler's List,"https://www.imdb.com/title/tt0108052/?ref=adv_li_tt","195 min","Biography, Drama, History"
6 The Lord of the Rings: The Return of the King,"https://www.imdb.com/title/tt0167260/?ref=adv_li_tt","201 min","Action, Adventure, Drama"
7 The Godfather Part II,"https://www.imdb.com/title/tt0071562/?ref=adv_li_tt","202 min","Crime, Drama"
8 12 Angry Men,"https://www.imdb.com/title/tt0050083/?ref=adv_li_tt","96 min","Crime, Drama"
9 Pulp Fiction,"https://www.imdb.com/title/tt0110912/?ref=adv_li_tt","154 min","Crime, Drama"
10 Inception,"https://www.imdb.com/title/tt1375666/?ref=adv_li_tt","148 min","Action, Adventure, Sci-Fi"
11 The Lord of the Rings: The Fellowship of the Ring,"https://www.imdb.com/title/tt0120737/?ref=adv_li_tt","178 min","Action, Adventure, Drama"
12 Fight Club,"https://www.imdb.com/title/tt0137523/?ref=adv_li_tt","139 min","Drama"
13 Forrest Gump,"https://www.imdb.com/title/tt0109830/?ref=adv_li_tt","142 min","Drama, Romance"
14 The Good, the Bad and the Ugly,"https://www.imdb.com/title/tt0060196/?ref=adv_li_tt","178 min","Adventure, Western"
15 The Lord of the Rings: The Two Towers,"https://www.imdb.com/title/tt0167261/?ref=adv_li_tt","179 min","Action, Adventure, Drama"
16 Jai Bhim,"https://www.imdb.com/title/tt15097216/?ref=adv_li_tt","164 min","Crime, Drama, Mystery"
17 Spider-Man: Across the Spider-Verse,"https://www.imdb.com/title/tt9362722/?ref=adv_li_tt","140 min","Animation, Action, Adventure"
18 Interstellar,"https://www.imdb.com/title/tt0816692/?ref=adv_li_tt","169 min","Adventure, Drama, Sci-Fi"
19 Goodfellas,"https://www.imdb.com/title/tt0099685/?ref=adv_li_tt","145 min","Biography, Crime, Drama"
20 The Matrix,"https://www.imdb.com/title/tt0133093/?ref=adv_li_tt","136 min","Action, Sci-Fi"
21 One Flew Over the Cuckoo's Nest,"https://www.imdb.com/title/tt0073486/?ref=adv_li_tt","133 min","Drama"
22 Star Wars: Episode V - The Empire Strikes Back,"https://www.imdb.com/title/tt0080684/?ref=adv_li_tt","124 min","Action, Adventure, Fantasy"
23 Oppenheimer,"https://www.imdb.com/title/tt15398776/?ref=adv_li_tt","180 min","Biography, Drama, History"
24 Se7en,"https://www.imdb.com/title/tt0114369/?ref=adv_li_tt","127 min","Crime, Drama, Mystery"
25 The Silence of the Lambs,"https://www.imdb.com/title/tt0102926/?ref=adv_li_tt","118 min","Crime, Drama, Thriller"
26 Saving Private Ryan,"https://www.imdb.com/title/tt0120815/?ref=adv_li_tt","169 min","Drama, War"
27 The Green Mile,"https://www.imdb.com/title/tt0120689/?ref=adv_li_tt","189 min","Crime, Drama, Fantasy"
28 Star Wars: Episode IV - A New Hope,"https://www.imdb.com/title/tt0076759/?ref=adv_li_tt","121 min","Action, Adventure, Fantasy"

```

Scraping w/ Parsehub: Preparing Your Data

The data you downloaded in CSV or (raw) Excel format may not seem very organized to look and further analyze in SPSS or R Studio.

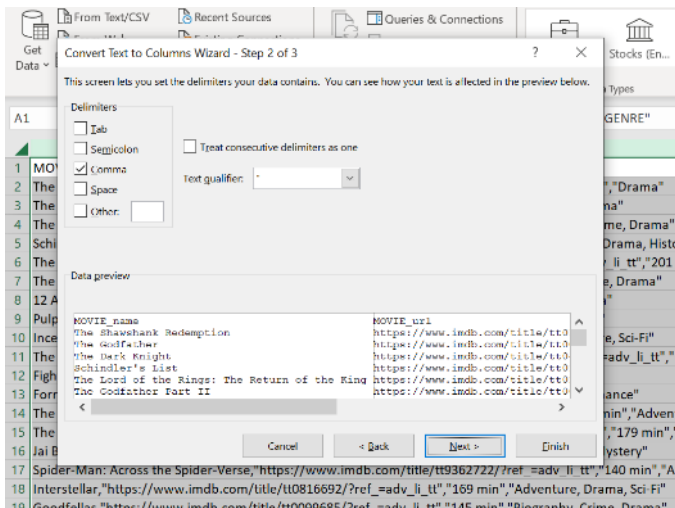
Then Use TEXT TO COLUMNS functionality in Excel to organize your data and make it more structured.

NOW your data is ready to pre-process (i.e. compute dummies etc) and further analysis.

您下载的 CSV 或 (原始) Excel 格式的数据可能看起来不够整齐，难以在 SPSS 或 R Studio 中进行进一步分析。

这时，请使用 Excel 中的“分列”功能整理数据，使其结构更加清晰。

现在，您的数据已准备好进行预处理（例如，计算虚拟变量等）和后续分析。



	A	B	C	D
1	MOVIE_name	MOVIE_url	MOVIE_DURATION	MOVIE_GENRE
2	The Shawshank Redemption	https://www.imdb.com/title/tt0111162	142 min	Drama
3	The Godfather	https://www.imdb.com/title/tt0011161	175 min	Crime, Drama
4	The Dark Knight	https://www.imdb.com/title/tt0468566	152 min	Action, Crime, Drama
5	Schindler's List	https://www.imdb.com/title/tt0111161	195 min	Biography, Drama, History
6	The Lord of the Rings: The Return of the King	https://www.imdb.com/title/tt0111161	201 min	Action, Adventure, Drama
7	The Godfather Part II	https://www.imdb.com/title/tt0011161	202 min	Crime, Drama
8	12 Angry Men	https://www.imdb.com/title/tt0011161	96 min	Crime, Drama
9	Pulp Fiction	https://www.imdb.com/title/tt0111161	154 min	Crime, Drama
10	Inception	https://www.imdb.com/title/tt1345836	148 min	Action, Adventure, Sci-Fi
11	The Lord of the Rings: The Fellowship of the Ring	https://www.imdb.com/title/tt0111161	178 min	Action, Adventure, Drama
12	Fight Club	https://www.imdb.com/title/tt0111161	139 min	Drama
13	Forrest Gump	https://www.imdb.com/title/tt0111161	142 min	Drama, Romance
14	The Lord of the Rings: The Two Towers	https://www.imdb.com/title/tt0111161	179 min	Action, Adventure, Drama
15	Jai Bhim	https://www.imdb.com/title/tt1516644	164 min	Crime, Drama, Mystery
16	Spider-Man: Across the Spider-Verse	https://www.imdb.com/title/tt9364756	140 min	Animation, Action, Adventure
17	Interstellar	https://www.imdb.com/title/tt0816692	169 min	Adventure, Drama, Sci-Fi
18	Goodfellas	https://www.imdb.com/title/tt0011161	145 min	Biography, Crime, Drama

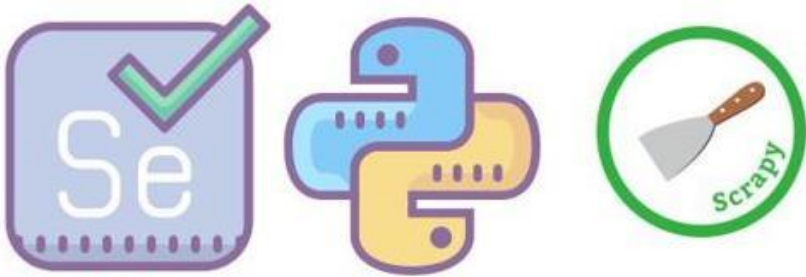
Scraping with Python – An Example

Optional for those who are interested. Not obligatory

Scraping with Python

PYTHON (environment) and its scraping related libraries are most popular for web scraping.

Most widely used general libraries: **Requests**, **Beautiful Soup**, **Scrapy** or **Selenium**. Those could be customized/used for different scraping projects.



Scraping with Python

Scraping-Data Retrieval with Python is such a broad area that can make a whole new course !

No worries: Because neither Python nor scraping with Python – are **NOT an obligatory part of this course.**

I suggest you can see this as a further path if you are deeply interested in scraping – so you can use in your thesis or later at your work,

For now since it is most widely used, general purpose scraping library I show you **an example with BeautifulSoup** to show it works:



[CLICK the Google Colab link for practicing yourself](#)

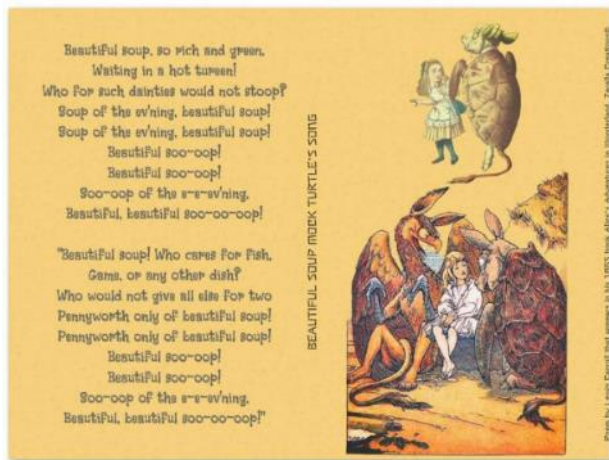
Scraping with Python - BeautifulSoup

Beautiful Soup is a Python package for parsing and scraping HTML based websites, including those with malformed markup.

Name comes from Beautiful Soup poem in Alice's Adventures in Wonderland and reference to term "tag soup" = poorly-structured HTML code.

It is often used by/with **Requests Python Library**. Requests is an HTTP client library for the Python programming language.

Requests is one of the most downloaded Python libraries, with over 300 million monthly downloads



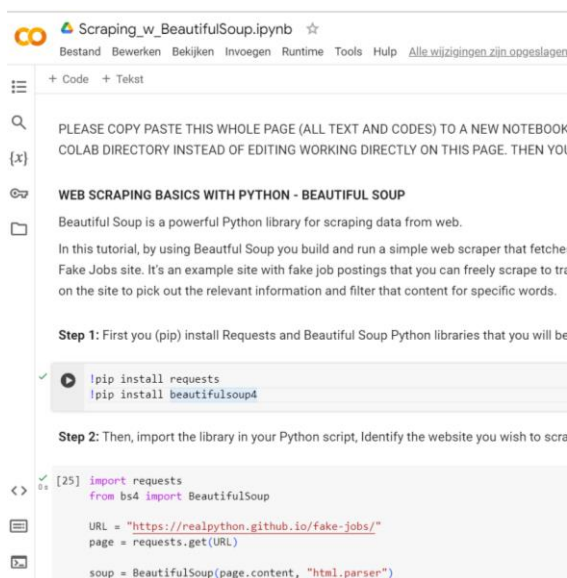
Web Scraping



Scraping with Python - BeautifulSoup

Please go to our COLAB (Beautiful Soup) scraping exercise by clicking the link below.

Just like in other COLAB exercises **copy paste this whole page** (all text and codes) to **a new notebook that you label and save in your google colab directory** instead of editing working directly on this page. then you can work on your own colab project.



Scraping_w_BeautifulSoup.ipynb

Bestand Bewerken Bekijken Invoegen Runtime Tools Hulp [Alle wijzigingen zijn opgeslagen](#)

+ Code + Tekst

PLEASE COPY PASTE THIS WHOLE PAGE (ALL TEXT AND CODES) TO A NEW NOTEBOOK COLAB DIRECTORY INSTEAD OF EDITING WORKING DIRECTLY ON THIS PAGE. THEN YOU CAN WORK ON YOUR OWN COLAB PROJECT.

WEB SCRAPING BASICS WITH PYTHON - BEAUTIFUL SOUP

Beautiful Soup is a powerful Python library for scraping data from web.

In this tutorial, by using BeautifulSoup you build and run a simple web scraper that fetches Fake Jobs site. It's an example site with fake job postings that you can freely scrape to try on the site to pick out the relevant information and filter that content for specific words.

Step 1: First you (pip) install Requests and BeautifulSoup Python libraries that you will be using.

```
!pip install requests
!pip install beautifulsoup4
```

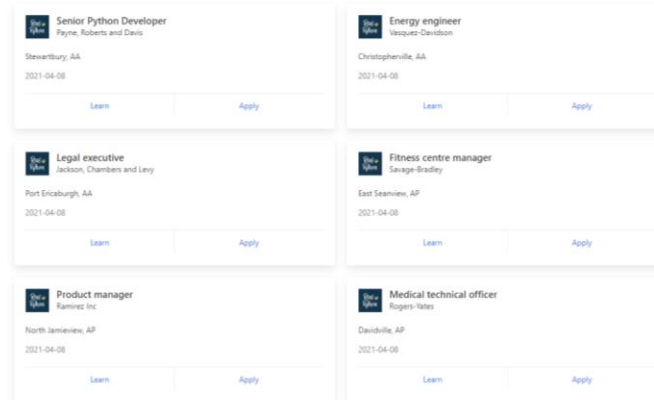
Step 2: Then, import the library in your Python script. Identify the website you wish to scrape.

```
[25] import requests
      from bs4 import BeautifulSoup

      URL = "https://realpython.github.io/fake-jobs/"
      page = requests.get(URL)

      soup = BeautifulSoup(page.content, "html.parser")
```

Fake Jobs for Your Web Scraping Journey



Senior Python Developer Wayne, Roberts and Davis Stewartbury, AA 2021-04-08 Learn Apply	Energy engineer Winquest-Devadason Christopherville, AA 2021-04-08 Learn Apply
Legal executive Jackson, Chambers and Leary Port Ercinburgh, AA 2021-04-08 Learn Apply	Fitness centre manager Savage-Bradley East Seaview, AP 2021-04-08 Learn Apply
Product manager Kamirez Inc North Jamiesville, AP 2021-04-08 Learn Apply	Medical technical officer Rogers-Rates Davidville, AP 2021-04-08 Learn Apply

NOW you will scrape from a fake job listing website – created for practice purposes for starting/ learning BeautifulSoup scraping process
<https://realpython.github.io/fake-jobs/>

Scraping with Python - BeautifulSoup



Step 1: First you (pip) install Requests and BeautifulSoup Python libraries that you will be using in scraping the website you wish to scrape.



```
!pip install requests  
!pip install beautifulsoup4
```



Step 2: Then, import the library in your Python script, Identify the website you wish to scrape and create a BeautifulSoup object:

```
[25] import requests  
      from bs4 import BeautifulSoup  
  
URL = "https://realpython.github.io/fake-jobs/"  
page = requests.get(URL)  
  
soup = BeautifulSoup(page.content, "html.parser")
```

First – you need to install Requests and BeautifulSoup libraries/packages to be able to start your scraping

Then you import BeautifulSoup from Requests library
You identify the website that you will scrape from

PLEASE NOTE

You need to have some basic understanding of HTML and how web pages are indexed/mapped by structuring different elements

If you have no information-background at all
Follow the steps on Colab exercise, read the explanations and try to understand how it works. The logic is very similar to what is done in Rvest.

Scraping with Python - BeautifulSoup

```
for job_element in job_elements:  
    title_element = job_element.find("h2", class_="title")  
    company_element = job_element.find("h3", class_="company")  
    location_element = job_element.find("p", class_="location")  
    print(title_element.text)  
    print(company_element.text)  
    print(location_element.text)  
    print()
```

Martinezburgh, AE

Surveyor, land/geomatics
Brown-Ortiz

Joshuatown, AE

Legal executive
Hartman PLC

West Ericstad, AA

Librarian, academic
Brooks Inc

Tuckertown, AE

Barrister
Washington-Castillo

Perezton, AE

After running all the steps (please try to read why we did each)
You will have scraped the JOBS LIST in the format given on the left.

Remember again – this is what we have scraped from the (fake) webpage listing different job positions below.

You don't have to learn PYTHON scraping in this course
But if you think of a career in digital marketing analytics it can be very useful for you in the future. Thus feel free to give a try!

Fake Jobs for Your Web Scraping Journey

 Senior Python Developer Payne, Roberts and Davis Stewartbury, AA 2021-04-08 Learn Apply	 Energy engineer Vasquez-Davidson Christopherville, AA 2021-04-08 Learn Apply
 Legal executive Jackson, Chambers and Levy Port Ericaburgh, AA 2021-04-08 Learn Apply	 Fitness centre manager Savage-Bradley East Seanview, AP 2021-04-08 Learn Apply

Scraping with Python: Further Tips



*There are specialized libraries such as `insta-scrapers`, `facebook-scrapers` etc. Note: META owned platforms such as Facebook-Instagram requires **authorization-permission** for scraping activity.*

评论类平台 (Review Platforms)

Review Platforms such as Fork, Yelp, Vivino etc.



You can scrape from these platforms by using Selenium, Scrapy, Requests (in Python) or if structure is eligible even with Data Miner.

购物或订房类平台 (Shopping / Booking Platforms)

Shopping or Booking platforms such as Booking.com, Bol.com, Amazon etc.



You can scrape from these platforms by using Selenium, Scrapy, Requests (in Python) or if structure is eligible even with Data Miner.

These platforms are often friendly or neutral towards scraping unless you use it for commercial purposes.

TIPS for Scraping Practice:

If you only need a small number of data points (tens to hundreds) it is always an option to do scraping **manually** thus: **copy-paste-organize**

That is often enough for your assignments. But note: Using tools such as Data Miner, R, Python:

1 Will make you save a lot of time

2 You can easily scrape large data sets (thousands, tens of thousands or even bigger)

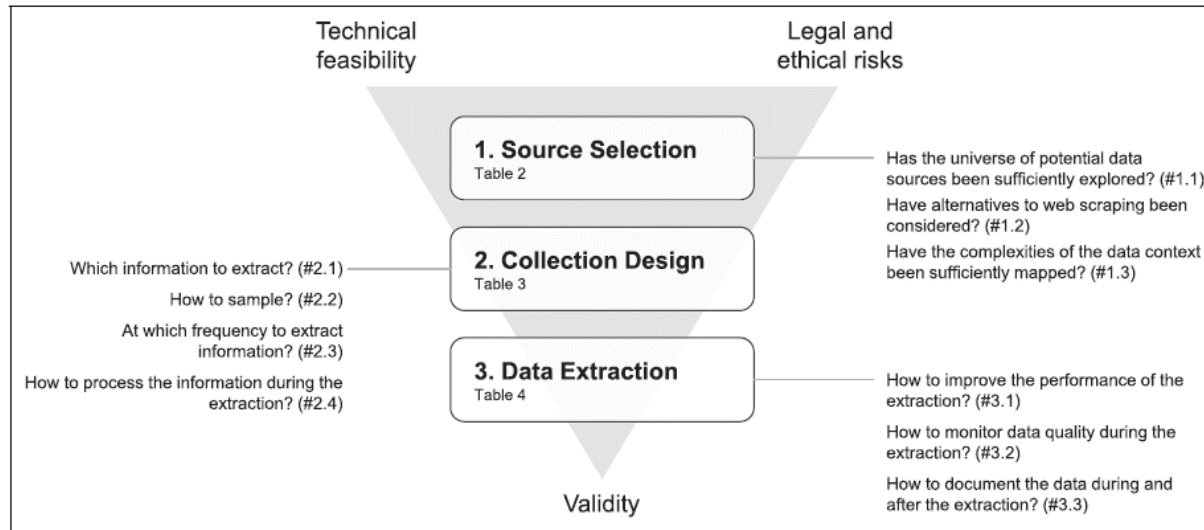
A Conceptual Framework for Collecting Web Data (Scraping)

Boegershausen et al. (2022) present a well-structured conceptual roadmap for scraping web data.

1 Assessing Technical feasibility & Legal/Ethical risks

2 Asking the right questions: What do we want to learn ? Sample size? What kind of data do we need for our analysis?

3 Monitoring the quality of the data and performance of scraping during and after the data collection



Scraping: Final Words - Practical Tips & Directions

- **Keep it SIMPLE!** Do NOT try to scrape everything on a page.
- Mostly a well-selected **3-5 column data** is sufficient for a useful analysis/ provides answers to interesting Marketing Questions.
- Think of your **marketing question** first and your simple analysis/model!
- For this course small scale use of **Parsehub** (or any of the methods we reviewed) is sufficient.
- Use R / Python libraries only if you are deeply interested in scraping.
- Some interesting ideas for exercising/practising: *What makes a hotel/restaurant/or flight more expensive ? What drive higher ratings/scores for restaurants/movies/products?*

The screenshot shows a hotel booking interface with several annotations in purple boxes:

- Display of choice**: Points to the top navigation bar.
- Narrowing down**: Points to the search filters on the left.
- Location value proposition**: Points to the search input field.
- Personalisation**: Points to the 'Filter by' section.
- Website value proposition**: Points to the headline 'Sydney: 203 properties found - including 66 value deals!'.
- Fear of missing out**: Points to the text 'Only 20% of properties are left on our site. Book soon to avoid missing out!'.
- Risk reduction**: Points to the 'Top picks for you' section.
- Social proof**: Points to the 'Popular now!' section.
- Sense of urgency**: Points to the 'Booked 26 times in the last 24 hours' text.
- Switching barrier**: Points to the 'See our last available rooms' button.

IF you find these too technical

YET you still enjoy getting data from the websites for your marketing projects &
IF you work with small data sets (few dozens or hundreds)

You can always start-do scraping by MANUAL copy-pasting. Yes it will take you a bit more time but in the end all scrapers (including humans) do the same scraping job in different ways :0)

Appendix 1

Scraping w/ Browser Extensions Data Miner

Optional. Not obligatory for the course.

Scraping with Data Miner



DATA MINER is a simple **CHROME** extension

It is a freeware extension that you can use/activate by clicking the **DATA MINER icon** on the upper-right corner

DATA MINER allows you to scrape web data by following simple steps – and creating your rules (recipes)

The image contains two screenshots of the Data Miner extension interface. The left screenshot shows the extension's popup menu on a webpage. A red circle labeled '1' highlights the 'Scrape This Page' button. The right screenshot shows the 'Page Scrape' configuration screen. A red circle labeled '2' highlights the 'Sandbox - Search' recipe, which is selected in the 'MY DATA' list. The interface includes tabs for 'Public Recipes', 'My Recipes', and 'Pinned'. The 'Public Recipes' list includes 'Sandbox - Search', 'Sandbox - URLs', and 'Sandbox - Detail'. The 'Page Scrape' screen shows options for 'Only recipes with Next Page Automation', 'Search by recipe name', and 'Scrape and only keep the result data of 1 scrape'. The 'MY DATA' section shows 'Saved Results 1' and 'Uploaded Files 0'. The 'Scraping Results' section shows a list of recipes and a 'Select a recipe from the left' prompt.

Scraping with Data Miner

Data Miner has ready to use recipes (pre-defined rules) that may already fit your data.

You can check the (ready) recipes and if one of them is sufficient for your needs – you can just RUN that recipe and collect your data.

However most recipes may not be tailored for your needs and **it is always good to customize (to decide)** which data you need.

I suggest you to create your own recipes in three simple steps – so that you can scrape what you really want to have as data

Data Miner interface showing the 'Page Scrape' screen. The 'Scraping Results' section displays a table with 4 rows of data. A red circle highlights the 'Scrape' button, and a red arrow points to it from a red circle containing the text 'Or Scrape and Append' and 'Or Next Page Automation'. A red circle with the number '4' is also present.

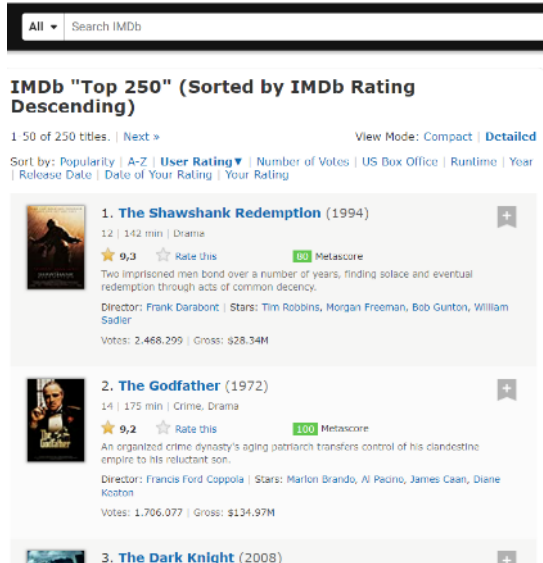
Data Miner interface showing the 'Page Scrape' screen. The 'Scraping Results' section displays a table with 4 rows of data. A red circle highlights the 'Download' button, and a red arrow points to it from a red circle containing the number '5'.

Some popular recipes are tested by Data Miner and are indicated by a “*” at the beginning of the recipe name. By using “tested” recipes you probably get better structured and more relevant data.

Data Miner: Creating Our Own Recipe to Scrape from Page

Let's make our own recipe with Data Miner by using **three basic steps**:

Let's say we would like to collect data on IMDB TOP 250 movies list: (a) name (b) IMDB rating score (b) meta score (d) genre of each movie



Let's start: We click Data Miner icon on top-right of our bar > Recipe Creator
Then we choose type of recipe: Here> List/Search page since we want to scrape the list of: TOP 250 Highly Rated movies from IMDB website.



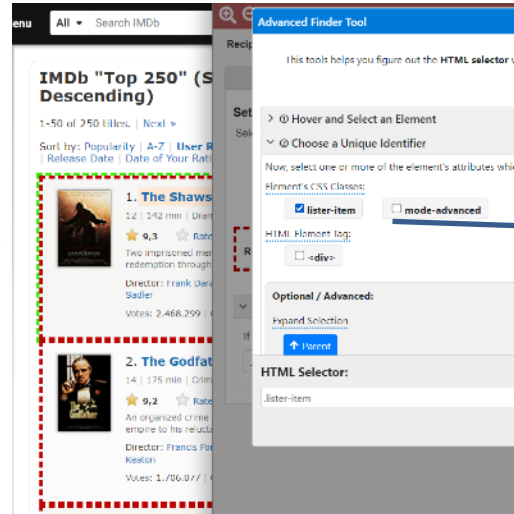
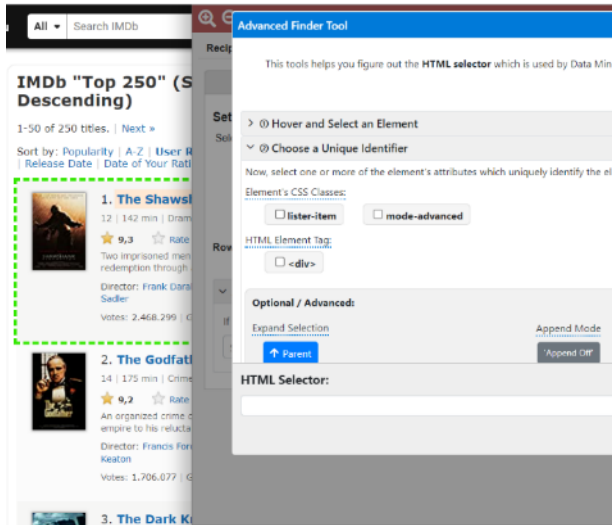
Data Miner Step 1: Defining the Container *(Each Row in your data)*

Our data will consist of various data on **250 MOVIES**. This means each row is represented by a movie and this repeats 250 times.

Then let's define each movie (thus: boxes related to each movie) as our main unit of analysis: thus our Container.



1a Place your mouse on the (whole) box relating to the first movie in the list
You will see that the box will be selected and will be turned into Orange in color
Then (while the box is selected) hit the SHIFT button and choose the box



1b When you choose the first box:
by checking/unchecking element classes
or HTML element types whether the
selection is correct

Here: check the 'list-item'
You will see then all movies (each box)
will be correctly selected as row elements
(in RED)

Data Miner: Step 2 *(Choose Sub-Elements: Specific Information You Need)*

Now that we want to collect data on **(a) name (b) IMDB rating score (b) duration in minutes; of each movie**

STEP 2.A: Next, we go to 5 COLS menu > and by “advanced finder” we select-define each data element one by one (each being a new column in your data).

STEP 2.B: We do a similar CONTROL-CHECK procedure to see whether the selection is correct by using Element Classes and HTML element type check boxes by checking-unchecking them and see whether the right element is selected.

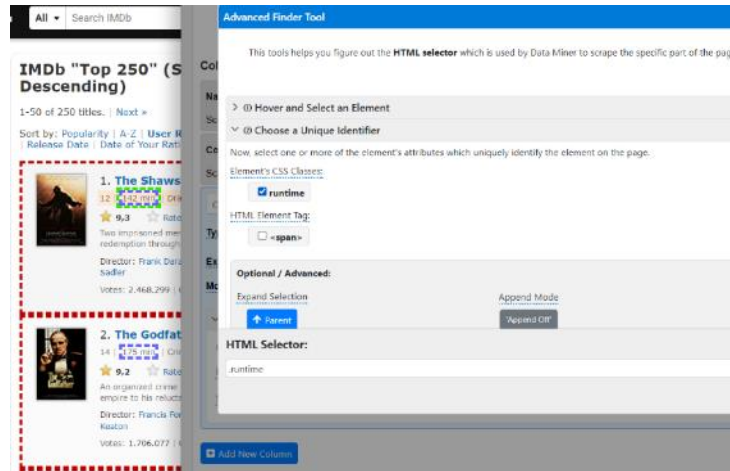
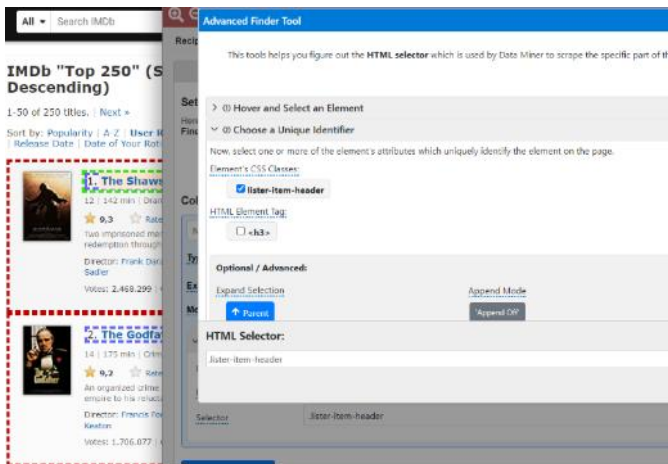


2 Check examples below:

Movie Name column is selected (element: lister-item-header)

Movie Duration column is selected (element: runtime)

Left
Right



Data Miner: Step 2 *(Choose Sub-Elements: Information You Need)*

Now you have all the elements-columns (variables) in your data-to-come defined: Name (b) IMDB rating score (b) Meta Score (d) Genre

So let's keep going to final step: STEP 3

Columns:

- Movie Name: 50 data
- IMDB Score: 50 data
- Runtime: 50 data

Type: ☒ Scrape ☐ Date ☐ Constant ☐ Skip

Extract: [Easy Column Finder](#) [More options](#)

Modify: [None: keep result as is](#)

[Advanced Settings](#)

[Add New Column](#)

Please Check the example:
We will have a data with three columns
1 Column: Movie Name
2 Column: IMDB Score
3 Column: Runtime

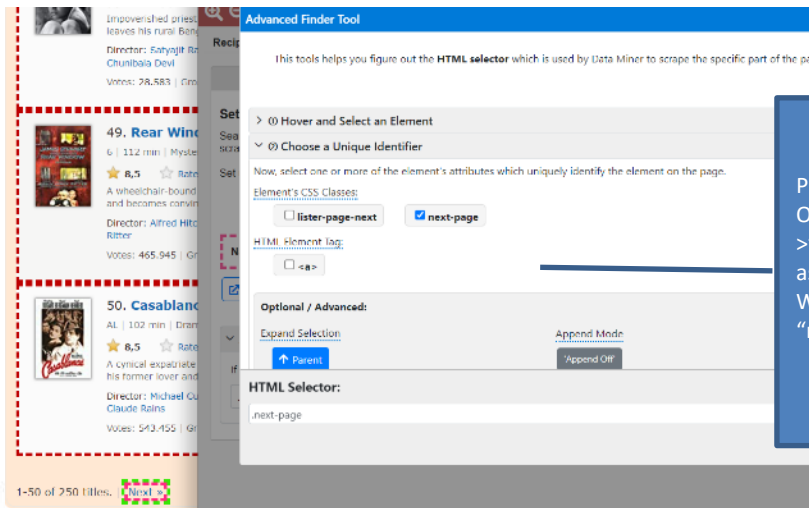
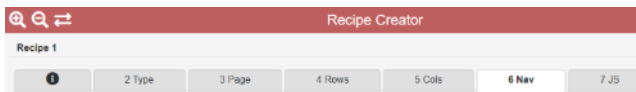
Data Miner: Step 3 *(Pagination > Tell the Scraper Not to Stop and to GO Next-Pages)*

There are 50 movies listed on each page – but in total there are 250 movies in IMDB TOP 250 list. Thus we have to scrape 5 pages to complete the list.

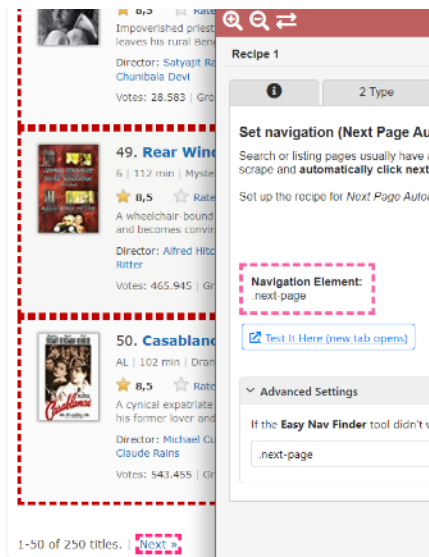
Thus we have to **TELL the scraper to automatically click (go) to NEXT page** to continue its work until it scrapes the last movie.

STEP 3.A: Next, we go to 6 NAV menu > and we select-define NEXT PAGE button; so that our scraper automatically goes to next page.

STEP 3.B: We do a similar CONTROL-CHECK procedure to see whether the selection is correct.



Please Check the example:
On (6 Nav – menu) we choose Let 'start "Next >" button at the bottom of the page as navigation element
We then check this by double-checking the "next-page" (CSS) element is selected



Data Miner: Running your Recipe (Scraping in Progress)

Now that we have completed all three steps (1) defining container <rows> (2) sub-elements <columns> (3) next-page automation.

Finally: we RUN the recipe and collect (scrape) our data and export in XLS format.

DONE: Now we have all the data about top 250 movies (IMDB) with their (a) name (b) IMDB score (c) meta score (d) genre, as we wanted.

Public Generic My Recipes

Recipes made by you or copied from other users.

☐ Hide recipes not working on this page

Search by recipe name

IMDB Top 250 Demo

Columns (3): Movie Name | IMDB Score | Runtime

☒ Selected ☐ Edit ☐ Delete ☐ Copy

Recipe At

Recipe 1

Columns (2): Column 1 | Column 2

Site: imdb.com

Recipe 1

Columns (3): Column 1 | Column 2...

Site: www.imdb.com

Recipe 1

Columns (3): Column 1 | Column 2...

Site: www.imdb.com

Recipe 1

Columns (3): Column 1 | Column 2...

Site: www.imdb.com

Scraping Results → **IMDB Top 250 Demo**

☐ New Scrape ☐ Scrape and Append ☒ Next Page Automation

Let Data Miner scrape and click next page button to continue scraping.

When you scrape with Next Page Automation, all scraped data accumulates into a single downloadable file.

Number of Pages to Scrape 5 pages

Delay between pages 8 seconds

Preview results for this recipe

This recipe scrapes 50 rows and 3 columns from this page

#	Movie Name	IMDB Score	Runtime
1.	The Shawshank Redemption (1994)	9,3	142 min
2.	The Godfather (1972)	9,2	175 min
3.	The Dark Knight (2008)	9,0	152 min

Please Check the example:
I also tell my scraper that I would like to use "Next Page Automation" that means I want to scrape 5 pages (in total 250 movies) and the information about those movies

logs Download

saved a few seconds ago 10/4/21, 2:19 PM **250 rows**

#	Movie Name	IMDB Score	Runtime
241.	Du rififi chez (1955)	8,1	118 min
242.	On the Waterfront (1954)	8,1	108 min

Then I run my recipe:
It lets me know when the scraping of 250 movies is done and my data is ready

Data Miner: Collecting Your Data

Then (when my recipe is done/finalized) – I collect my data. Here I choose to save it in Excel format. You can also choose different formats.

It is easy and fast (you will realize) as you do some practice. Please check the data below:

Movie Name		
A	B	C
Movie Name	IMDB Score	Runtime
1. The Shawshank Redemption (1994)	9,3	142 min
2. The Godfather (1972)	9,2	175 min
3. The Dark Knight (2008)	9,0	152 min
4. The Godfather: Part II (1974)	9,0	202 min
5. De 12 gezworenen (1957)	9,0	96 min
6. The Lord of the Rings: The Return of the King (2003)	8,9	201 min
7. Pulp Fiction (1994)	8,9	154 min
8. Schindler's List (1993)	8,9	195 min
9. Inception (2010)	8,8	148 min
10. Fight Club (1999)	8,8	139 min
11. The Lord of the Rings: The Fellowship of the Ring (2001)	8,8	178 min
12. Forrest Gump (1994)	8,8	142 min
13. The Good, the Bad and the Ugly (1966)	8,8	161 min
14. The Lord of the Rings: The Two Towers (2002)	8,7	179 min
15. The Matrix (1999)	8,7	136 min
16. Goodfellas (1990)	8,7	146 min
17. Star Wars: Episode V - The Empire Strikes Back (1980)	8,7	124 min
18. One Flew Over the Cuckoo's Nest (1975)	8,7	133 min
19. Gisaengchung (2019)	8,6	132 min
20. Interstellar (2014)	8,6	169 min
21. Cidade de Deus (2002)	8,6	130 min
22. Spirited Away (2001)	8,6	125 min
23. Saving Private Ryan (1998)	8,6	169 min
24. The Green Mile (2009)	8,6	129 min

MY SCRAPED DATASET BASED ON MY RECIPE

1 Each **row** represents a different movie (in total 250 rows)

2 I have a **column** for: name, IMDB score and runtime

Now you can try/practice with other website. Suggestions:

Booking.com hotel listings

Yelp (or Fork) restaurant reviews

Bol.com or Coolblue product listings

Appendix 2: **Scraping with R (rvest)**

Optional for R Users. Not obligatory

Web Scraping with R (Rvest)

R can be used for advanced level scraping (very large datasets, complex multi-layer websites) [RVEST](#) is the most commonly used package for scraping

However: You will need basic **coding and HTML-CSS knowledge** to scrape with R.

Not required in this course. But if you want to increase your level of expertise in scraping you can give a try !

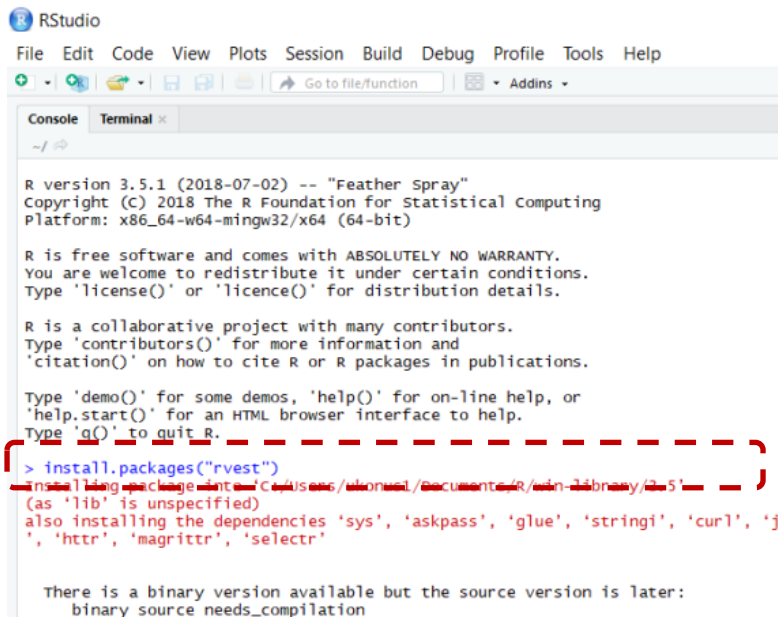


Scraping with R (rvest): Installation

[rvest](#) is a powerful R package to scrape web data in different formats/ larger quantities (completely free)

Use R console (command screen) to install **rvest**. If you are less familiar with R you may wish to use **Rstudio** interface.

Install the package by using the command line: `install.packages("rvest")`



The screenshot shows the RStudio application window. The title bar says "RStudio". The menu bar includes File, Edit, Code, View, Plots, Session, Build, Debug, Profile, Tools, and Help. Below the menu bar is a toolbar with icons for running, saving, and other functions. The main pane is split into two tabs: "Console" and "Terminal". The "Console" tab is active, showing the R version 3.5.1 (2018-07-02) -- "Feather Spray" and the copyright notice. The user has entered the command `> install.packages("rvest")` in the console. The output shows that the package is being installed from source, and the dependencies 'sys', 'askpass', 'glue', 'stringi', 'curl', 'j', 'httr', 'magrittr', and 'selectr' are also being installed. The console also displays the message "There is a binary version available but the source version is later: binary source needs_compilation".

```
R version 3.5.1 (2018-07-02) -- "Feather Spray"
Copyright (c) 2018 The R Foundation for Statistical Computing
Platform: x86_64-w64-mingw32/x64 (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under certain conditions.
Type 'license()' or 'licence()' for distribution details.

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' on how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> install.packages("rvest")
Installing package into 'C:/Users/ukonai/Documents/R/win-library/3.5'
(as 'lib' is unspecified)
also installing the dependencies 'sys', 'askpass', 'glue', 'stringi', 'curl', 'j',
'httr', 'magrittr', 'selectr'

There is a binary version available but the source version is later:
binary source needs_compilation
```

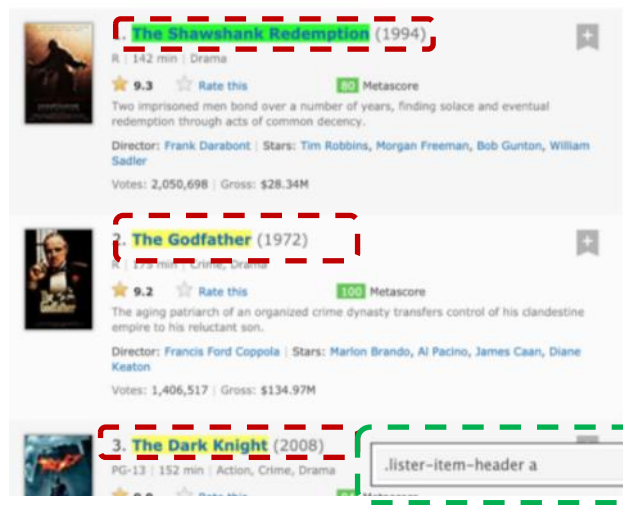
Scraping with R (rvest): *SelectorGadget* > to spot-define CSS sub-elements

To start scraping: Now you need to install **SelectorGadget** (Chrome Extension) on your Google Chrome.

SelectorGadget is a simple tool (used with rvest) to spot, mark and define **CSS** (information elements: such as keywords, ratings) on websites.

you find these elements on HTML websites with **SelectorGadget** and then **show (define) them in rvest** > so; which elements to scrape

How to use Selector Gadget ?



What is CSS – CSS Selectors?

CSS (elements) describes which/ how HTML elements in the web page should be displayed.

Each element you see on the list has a CSS in HTML code
For example: Title of each movie = .list-item-header

CSS selectors are used to "find" (or select) the HTML elements you want to style.

HERE YOU FIND THE CSS ELEMENT OF "MOVIE TITLE"
BY USING SELECTOR GADGET
IT AUTOMATICALLY SHOWS BELOW:
HERE: ".list-item-header a"
WHEN YOU CLICK THE MOVIE TITLES

Scraping with R (rvest): Defining Website (List) and Scraping the Elements

Then I go back to R command screen. I load the rvest library + define the web site I would like to scrape from with two simple lines.

Here I would like to scrape from IMDB TOP 250 movies list. You can scrape from anywhere you want.

```
> library('rvest')
> url <- 'https://www.imdb.com/list/ls041322734/'
> webpage <- read_html(url)
>
> movie_title_data_html <- html_nodes(webpage, '.list-item-header')
> #converting the movie title data to text
> movie_title_data <- html_text(movie_title_data_html)
> #Let's have a look at the top 250 movie titles
> head(movie_title_data)
```

→ STEP 1

→ STEP 2

→ STEP 3

[1]	"\n	1.\n	\n	The Shawshank Redemption\n	(1994)\n"	"\n	2.\n	\n	The Godfather\n	(1972)\n"
[3]	"\n	3.\n	\n	The Godfather: Part II\n	(1974)\n"	"\n	4.\n	\n	The Dark Knight\n	(2008)\n"
[5]	"\n	5.\n	\n	12 Angry Men\n	(1957)\n"	"\n	6.\n	\n	Schindler's List\n	(1993)\n"

STEP 1: I load my **rvest** package and I specify the weblink I want to scrape (here it is IMDB TOP 250 movies list)

STEP 2: I then go to website and with Gadget Selector I spot what information about movies I would like to scrape (see previous slide)

STEP 3: Finally I extract (scrape) movie title data from the website, convert it into text format and display the data !

步骤 1: 我加载 rvest 包, 并指定要抓取的网页链接 (这里是 IMDB TOP 250 电影榜单)。

步骤 2: 然后我访问该网站, 并使用 Gadget Selector 选择要抓取的电影信息 (参见上一张幻灯片)。

步骤 3: 最后, 我从网站提取 (抓取) 电影标题数据, 将其转换为文本格式并显示出来!

Thank You